Climate change and climate change impacts on São Tomé and Príncipe with a focus on fishery and tourism sectors in a Blue Economy context

Fact Sheet - São Tomé and Príncipe

Climate Change Impacts and Risks

São Tomé and Príncipe is a Small Island Developing State (SIDS) comprised of two islands in the Gulf of Guinea. São Tomé and Príncipe is heavily dependent on its blue economy, particularly on fisheries and tourism, and has an exclusive economic zone (EEZ) ~160 times larger than its land mass.

CLIMATE AND OCEAN HAZARDS

CLIMATE

Historical changes

Dataset

W5E5 bias-corrected reanalysis dataset

Temperature trends

Surface air temperatures have increased by ~0.01-0.015°C per year in São Tomé and Príncipe between 1980-2020, with significant increases in October-December.

Precipitation trends

While precipitation in São Tomé and Príncipe has decreased in January-March months (~ 0.3 mm/year), it has increased in October-December months (~ 3 mm/year). Precipitation in Príncipe has increased **2 times** compared to that in São Tomé.

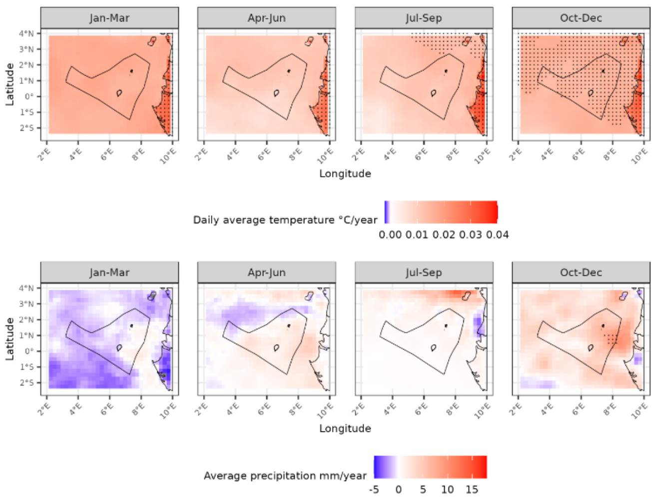

Figure 1. *Trend analysis for historical annual mean surface temperature (top) and annual mean precipitation (bottom). Black dots indicate values that are statistically significant.*

Future changes

Dataset

Three GCMs (General Circulation Models), namely MOHC-HadGEM2-ES, MPMI-M-MPI-ESM-LR, and NCC-NorESM1-M dynamically downscaled by the CORDEX-CORE initiative using two (Regional Climate Model) REMO2015 and RegCM4 resulting in six available simulations. Temperature and precipitation trends were determined for two future climate scenarios (RCP 2.6 & 8.5) and time periods (2020-2049 & 2050-2079).

Temperature trends

Average surface temperatures were projected **to increase by ~0.8-1°C** compared to baseline period for both 2020-2049 & 2050-2079 under RCP 2.6. Under RCP 8.5, temperatures were projected to **increase by ~1°C by 2020-2049 and by ~2.3°C by 2050-2079**.

Precipitation trends

In São Tomé, average precipitation was projected to increase by ~30 mm/year in April-June and October-December by 2020-2049 under RCP 2.6 as well as by both time periods under RCP 8.5. However, precipitation was projected to decrease in January-March. Similar findings were observed in Príncipe although precipitation increases under RCP 8.5 were **2 times** those in São Tomé.

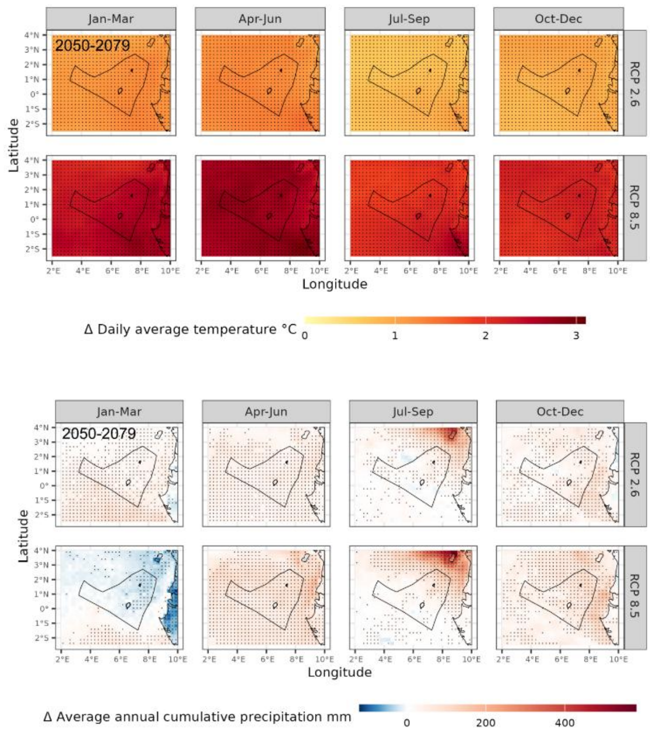

Figure 2. *Trend analysis for future projected annual mean surface temperature (top) and annual mean precipitation (bottom). Black dots indicate at least 80% model agreement on sign of change.*

OCEAN

Historical changes

Datasets

COPERNICUS marine data store for analysis of ocean variables. Datasets used: Global Ocean ODYSSEA L4 Sea Surface Temperature (temperature), Global Ocean Ensemble Physics Reanalysis (salinity, mixed layer depth, bathymetry), Global Ocean Biogeochemistry Hindcast (O2, primary productivity, pH).

Sea surface temperature trends

The São Tomé and Príncipe exclusive economic zone (EEZ) has warmed by ~0.015-0.02°C per year between 1993-2023. Notably, ocean warming is significantly slower close to the island coastlines compared to open ocean.

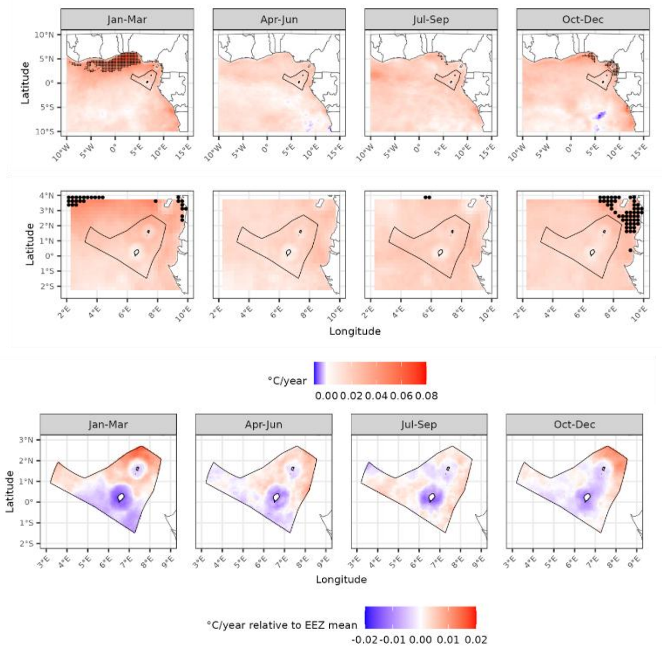

Figure 3. *Trend analysis for historical annual mean sea surface temperature in the Gulf of Guinea and EEZ (top). Bottom plots show annual mean sea surface temperature trend relative to the EEZ average where blue colour shows regions warming at slower rate than EEZ average.*

Other trends

In addition to sea surface temperature, we observed significant trends of decreasing ocean pH (acidification), deoxygenation, and salinity reduction.

Future changes

Datasets

Three Earth System Models (GFDL-ESM4, MPI-ESM1-2-HR, and UKESM1-0-LL) and one model ensemble (IPSL-CM6A-LR). Models were run for two future climate scenarios (SSP1 2.6 & SSP5 8.5) and time periods (2020-2049 & 2050-2079).

Sea surface temperature trends

Average sea surface temperatures are projected to **increase by 0.6-1°C** across both 2020-2049 & 2050-2079 under an SSP1 2.6 scenario. Under SSP5 8.5, temperatures are projected to **increase by ~0.8°C by 2020-2049 and by ~1.8°C by 2050-2070.**

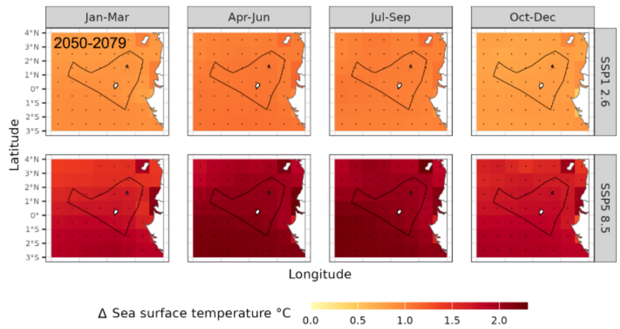

Figure 4. *Trend analysis for future projected annual mean sea surface temperature in the Gulf of Guinea and EEZ. Black dots indicate at least 80% model agreement on sign of change*

Other trends

Similar to historical analyses, climate models projected decreases in ocean pH (acidification) and oxygen levels.

CLIMATE EXPOSURE AND VULNERABILITY

Like many SIDS, São Tomé and Príncipe is largely reliant on environmental resources for its economy and food security. Due to São Tomé and Príncipe's sandy beaches and high biodiversity, tourism is an increasingly important part of the economy. Tourism indirectly contributes to 14% of GDP, 10% of employment, and makes up ~74% of overall exports. Fisheries are the main source of income for coastal communities in São Tomé and Príncipe and fish makes up ~50% of the inhabitants' animal protein intake.

The primary vulnerabilities of tourism and fisheries are expected to come from coastal erosion, extreme weather events, and decreasing biodiversity. Coastal erosion will impact São Tomé and Príncipe's sun and beach tourism by eroding beaches and disrupting coastal infrastructure such as hotels. Coastal erosion may also impact coastal fishing communities who may need to re-locate to less impacted areas. Changing climates may also lead to reductions in biodiversity. Eco-tourism is a major tourism opportunity for São Tomé and Príncipe and may be at risk from climate change. Further, marine biodiversity is essential for fisheries to ensure consistent catches and minimise the risk of fish population collapse.

ADAPTIVE CAPACITY

The current adaptive capacity of São Tomé and Príncipe to respond to climate change is low. For tourism, adaptive capacity is limited due to a lack of international marketing, underdeveloped public transportation, and insufficient regulation of natural resource exploitation (e.g sand, large trees). Insufficient regulations also limit adaptive capacity for fisheries with high risks of over-fishing and depletion of fish stocks. A significant challenge also lies in the high poverty rates (65% below poverty line). As poverty becomes more prevalent with deteriorating tourism and fisheries, the enforcement of resource exploitation regulations becomes more difficult.

PROJECTED CLIMATE CHANGE AND OCEAN CHANGES – EXECUTIVE SUMMARY

Table 1. Climate summary made considering distant future (2050-2079) relative to baseline (1980-2005). Confidence is based on agreement between the six CORDEX-CORE simulations used in this document

Variable	**RCP 2.6**	**RCP 8.5**	**Confidence**
Temperature	~ 1°C	~2.5°C	High
Precipitation	~ 20 mm/year	~20 mm/year	Medium
Unusual events (heatwaves)	+ 2-4 events/year	+ 2-4 events/year	High
Unusual events (dry days)	+- 1-2 days/year	+- 1-5 days/year	Low
Unusual events (high precipitation)	+ 0-0.5 days/year	+ 0-1 days/year	Low

Table 2. Ocean summary made considering distant future (2050-2079) relative to baseline (1980-2005). Confidence is based on agreement between three Earth System Models and one ensemble model.

Variable	**SSP1 2.6**	**SSP5 8.5**	**Confidence**
Sea surface temperature	1°C	2°C	High
Sea surface pH	-0.1	-0.2	High
Sea surface oxygen	-0.003 mol/m^3	-0.006 mol/m^3	High
Unusual events (marine heatwave)	8 times	8 times	High
Unusual events (low pH)	10 times	10 times	High
Unusual events (low oxygen)	15 times	20 times	High

IMPACTS

Sea level rise

Sea levels around São Tomé and Príncipe were found to have increased by ~0.003m per year between 1993-2023, amounting to a **total increase of ~0.09m**.

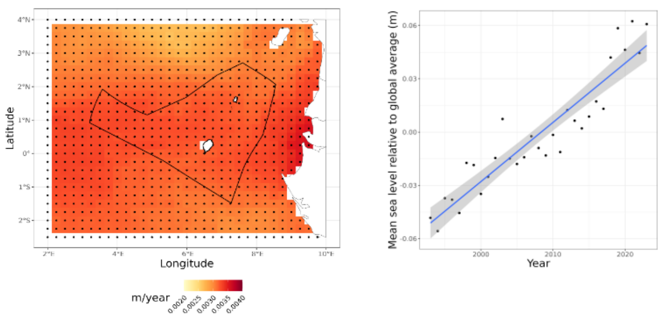

Figure 5. Historical changes in sea level in the EEZ between 1993-2023. Data from Copernicus Global Ocean Ensemble Physics Reanalysis dataset.

Fish biomass impacts

The impact of climate change on the biomass of demersals and pelagics of different sizes (small, medium, and large) was determined using biomass impact models from ISIMIP run with CMIP6 future climate projections.

Future climate change is projected to reduce the biomass of both demersals and pelagics of all sizes. The biggest decreases were projected for large and small pelagics. While there was high model agreement on the extent of biomass loss for small pelagics, there was high disagreement for large pelagics with some models projecting large decreases. Small pelagics are the predominant fish type sold and eaten by coastal communities in São Tomé and Príncipe.

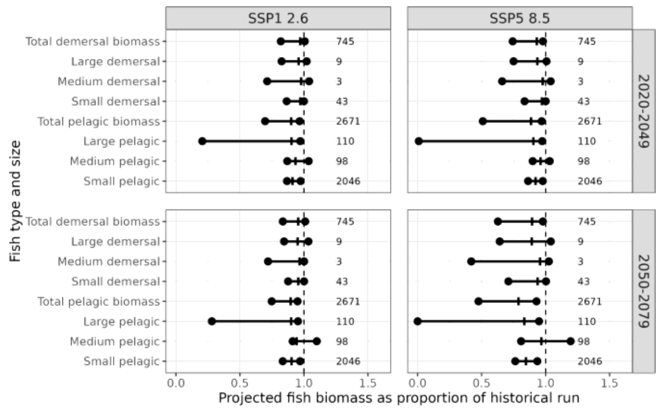

Figure 6. Projected fish biomass change based on impact and climate model ensemble. Vertical lines show model median while dots and horizontal lines show 17th and 83rd percentiles. Numbers to right show projected present day yields.

Fish species occurrence impacts

Climate change impacts on fisheries were further assessed through species distribution modelling using the machine-learning approach MaxEnt. This method allowed the generation of occurrence probabilities for 60 species identified as important for blue economy.

Under future climatic conditions (SSP5 8.5), the probability of species occurrence was projected to **increase for 10 species**, **remain unchanged for 14 species**, and **decrease for 36 species**. Geospatial analysis highlighted geospatial trends whereby climate impacts were reduced to the east of São Tomé and Príncipe and close to the coastlines. This latter finding is supported by historical ocean analyses which showed that ocean warming is buffered close to coastlines.

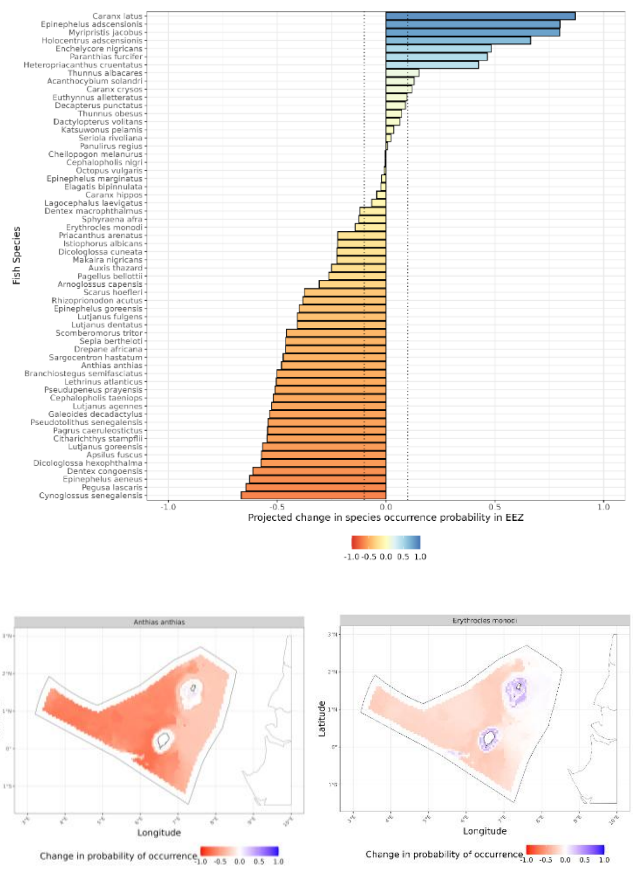

Figure 7. *Projected change in species occurrence probabilities under SSP5 8.5 scenario in EEZ (top). Bottom shows geospatial distribution of occurrence probabilities in EEZ for two species with climate impact buffering to east of EEZ and close to coastlines.*

Climate impact on fish distributions was assessed further for 15 species identified as candidates for sports fishing. Decreases in species occurrence probability were projected for 7/15 species. No change was projected for 5/15 species and minor increases were projected for 3/15 species.

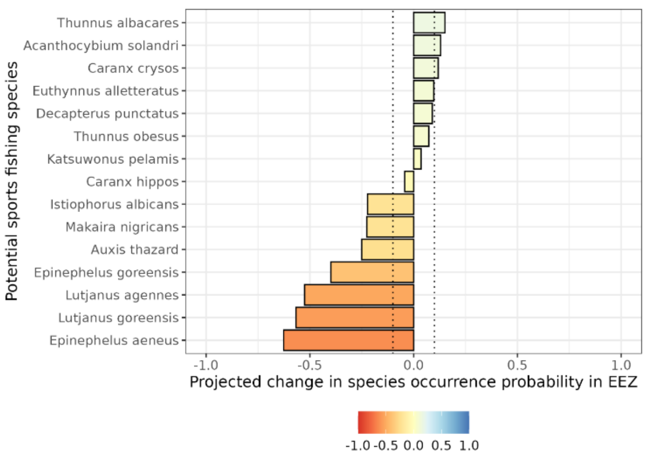

Figure 8. *Projected change in occurrence probabilities of 15 target sports fishing species under SSP5 8.5 scenario in EEZ.*

PRELIMINARY ADAPTATION RECOMMENDATIONS

To minimise the negative impacts of climate change, we have prepared the following list of preliminary adaptation recommendations

<u>Tourism</u>

In support of the recent WACA+ project, we also recommend:

- Construction of dikes, breakwaters (e.g gabions), and protective walls in most vulnerable areas.
- Creating safe expansion zones for coastal communities.
- Establishment of a climate alert system on the state of the sea
- Rehabilitation of tide stations
- Sand dredging to repair eroded beaches
- Diversification of activities and support measures for populations whose livelihoods depend on coastal activities
- Construction of drainage ditches and water tanks

We also recommend further efforts to promote sustainable eco-tourism including:

- Monitoring and cultivating healthy coral reefs with inclusion of artificial reefs where necessary.
- Mangrove restoration activities either for tourists or funded by mangrove tourism.
- Development of sports fishing industry
    - Diversify job opportunities (fishing instructors, ship captains)
    - Improved awareness of conservation necessity through catch and release

<u>Fisheries</u>

- Enforcement and further development of marine protected areas beyond the eight recently announced.
- Diversification of fishing activities through movement from artisanal fishing to industrialisation
    - Increase job security by providing non-fishing jobs in industry
    - Improve accountability for abiding by fishing regulations
- Development and enforcement of legal fishing practices:
    - Returning juveniles
    - Catch limits
    - Limits on number of fishing vessels
    - Use of unweighted purse seine fishing methods
- Development of key infrastructure within fishing communities including:
    - Additional fishing ports
    - Refrigeration units for catch storage
    - Manufacturing units for building fishing materials
    - Workshop for building fishing boats

CLIMATE MODEL DOWNSCALING

Approach

Bias-corrected CORDEX-CORE ensemble climate model biases were determined for cumulative precipitation, heavy rainfall (>20mm) day frequency, and dry day frequency. We then compared model biases of CORDEX-CORE historical run against the historical run of an independent statistically downscaled, bias-corrected climate model ensemble.

Results

Model biases were generally found to be low for both the CORDEX-CORE and statistically downscaled ensembles. Notable deviations in model biases between the ensembles occurred in Principe where dry days were over-estimated by the CORDEX-CORE ensemble and rainfall was over-estimated for a single year. Model biases were sufficiently low that we were confident moving forward with bias-corrected CORDEX-CORE models for climate projection analysis.

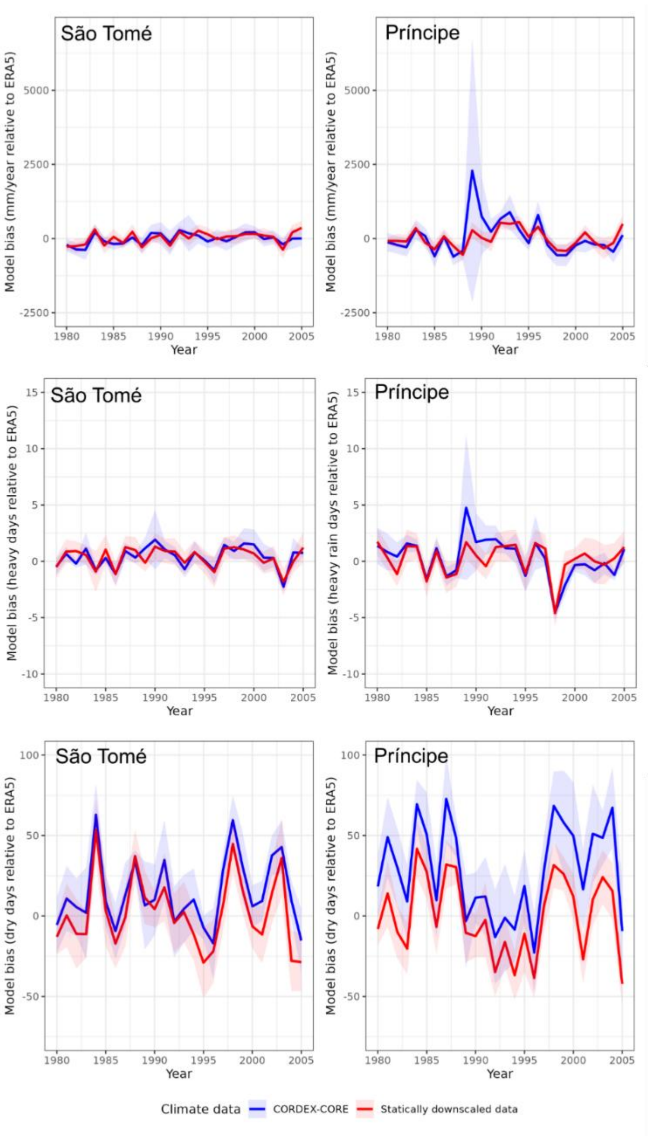

Figure 9. *Model biases of bias-corrected CORDEX-CORE and independent statistically downscaled climate dataset. Models biases are determined against the ERA5 re-analysis dataset. Model biases were calculated for cumulative precipitation (top plots), heavy rainfall day frequency (middle plots), and dry day frequency (bottom plots).*